

Problem 29

A body is in free fall from a height h . At point A, it has a velocity of $v_A = 29.43$ m/s, and at point B, its velocity is $v_B = 49.05$ m/s. Determine the height difference between points A and B. Also, find the time required for the body to travel from A to B.

Solution

The velocity of a freely falling body is given by the kinematic equation*:

$$v^2 = v_0^2 + 2gh.$$

Since the body is in free fall, the initial velocity at point A is v_A , and the velocity at point B is v_B . The height difference h_{AB} is given by:

$$v_B^2 = v_A^2 + 2gh_{AB}.$$

Substituting the given values:

$$(49.05)^2 = (29.43)^2 + 2(9.81)h_{AB}.$$

Calculating:

$$2405.90 = 866.14 + 19.62h_{AB},$$

$$1539.76 = 19.62h_{AB},$$

$$h_{AB} \approx 78.5 \text{ m}.$$

Now, using the kinematic equation* for time:

$$v_B = v_A + gt,$$

solving for t :

$$t = \frac{v_B - v_A}{g} = \frac{49.05 - 29.43}{9.81}.$$

Calculating:

$$t \approx \frac{19.62}{9.81} = 2 \text{ s}.$$

Thus, the height difference between points A and B is approximately 78.5 m, and the time required to travel this distance is 2 s.